Abnormal muscle tone

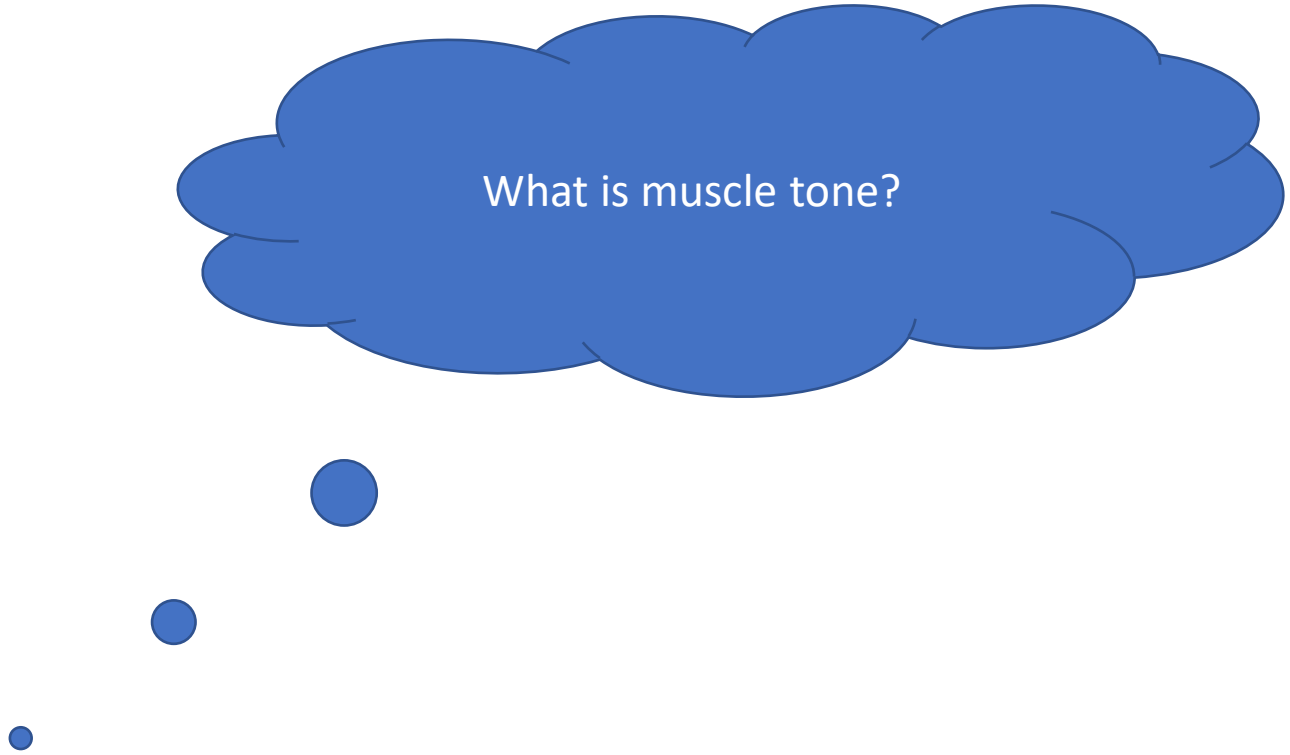
Prof Isaac Sorinola

Learning Outcomes

By the end of this lecture, students should be able to:

- Describe and recognize the clinical presentation of muscle tone impairments such as:
  - Hypertonia & hypotonia
  - Rigidity
  - Involuntary movements
- Identify neural and non-neural factors in spastic hypertonia
- Understand the aims of physiotherapy intervention in the management of spastic hypertonia

What is muscle tone?



What is muscle tone?

- When the clinician passively moves a joint in a normal, relaxed limb, a certain resistance, known as tone, is detected.
  - inertia of the limb
  - viscoelastic properties of the muscle and joint
  - No neural response to stretch (Burke, 1998)



Hypotonia (↓ resistance)

- resistance decreases whenever the reflex arc is damaged (usually at the level of the peripheral motor or sensory nerve), but it may also decrease with primary muscle or spinal cord disease.

Hypertonia (↑ resistance)

- increase in resistance to stretch in one direction. It often occurs with the presence of a lesion of the upper motor neurons anywhere along the pathway from the motor cortex to the ventral horn neurons in the spinal cord. There may be neural and non-neural contributory impairments

Hypertonia - ↑ resistance to passive stretch

Rigidity

1. Present in both extensor and flexor muscle groups
2. Not velocity dependent
3. Normal deep tendon reflexes (++)
4. Lead pipe &/or cogwheel



Spasticity

1. Usually present in one muscle more than the opposing antagonist muscle .e.g., biceps > triceps
2. Velocity dependent
3. Associated with exaggerated deep tendon reflexes (+++/+)
4. Associated with characteristic postures (UL flexor, LL Extensor)

Rigidity in Parkinson's Disease

Consequences of rigidity

- stiff and/or inflexible muscles
- reduced facial expression
- stooped posture
- pain and muscle cramps
- difficulty turning when in bed, walking or during transfers
- reduced arm swing when walking
- difficulty with everyday activities

Pharmacological management

- Levadopa
- Dopamine agonists
- Amantadine
- Selegiline



Physiotherapy

- Maintain range of movement
- Maintain mobility
- Maintain function

Upper Motor Neurone Syndrome

NEGATIVE FEATURES

Muscle weakness
Fatigue
Impaired co-ordination
Impaired motor control
Impaired motor planning

POSITIVE FEATURES

Spasticity/Rigidity
Hyper-reflexia
Babinski response
Abnormal movements
Athetosis/Clonus/Dystonia

Features of upper motor neurone syndrome

Positive Features	Afferent drive: in response to peripheral stimulation	
Additional or exaggerated phenomena	Proprioceptive stretch reflexes	
		Spasticity Clonus Tendon hyperreflexia with irradiation Positive support reaction
	Nociceptive reflexes:	
		Flexor spasms
	Cutaneous reflexes	
		Extensor spasms Extensor plantar response (positive Babinski)
	Efferent drive: Supraspinal activity: observed during movement	
		Spastic dystonia Associated reactions
	Disordered control of voluntary movement:	
		Reduced reciprocal inhibition leading to pathological (spastic) co-contraction Excessive reciprocal inhibition leading to apparent weakness

Features of upper motor neurone syndrome

Negative Features		
Loss or reduction of phenomena		
		Weakness Fatiguability Loss of dexterity Acute hypotonia
Adaptive Features		
Soft tissues	Non-neural; contributions (biomechanical component)	Stiffness and shortening of peripheral soft tissues structures
	Neural contributions (overactive muscle contraction)	
		Spasticity
		Spastic dystonia
		Hyperreflexia

Spasticity – what is Spasticity?

- “a velocity-dependent increase in tonic stretch reflexes with exaggerated tendon jerks resulting from hyperexcitability of the stretch reflex as one component of the upper motor neuron syndrome (Lance, 1980)



In addition, spasticity may be associated with:

- Characteristic postures
- Primitive reflexes
- Co-contraction
- Stereotyped movement patterns
- Associated movements



Common patterns of spasticity after stroke



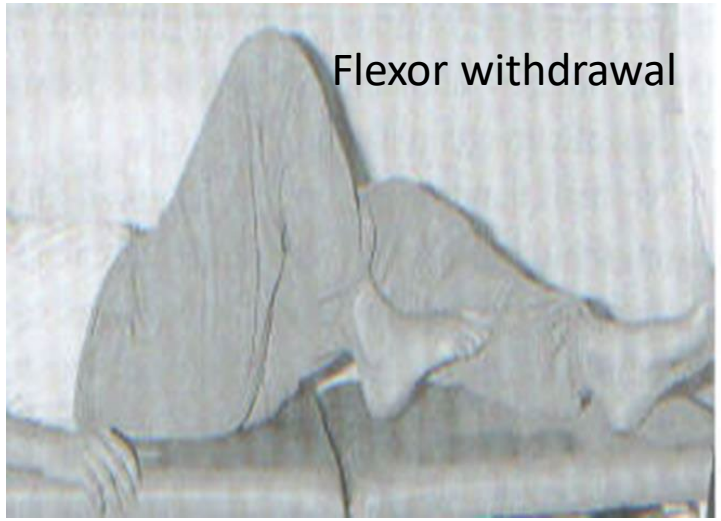
Shoulder girdle retraction, adduction, medial rotation, Elbow and forearm flexion, pronation, Wrist and hand flexion



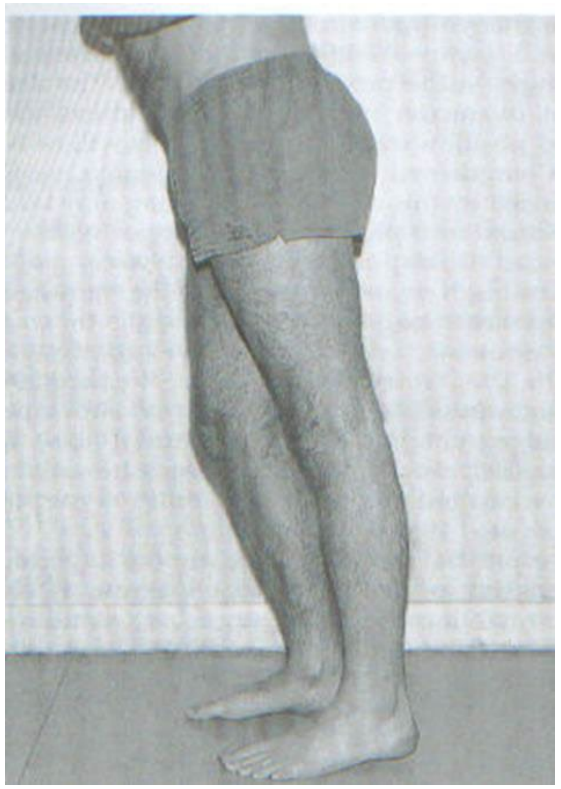
Hip internal rotation, adduction, extension
Knee extension, Ankle inversion, plantar flexion

Individual pattern determined by neural factors AND non-neural factors associated with disuse induced muscle shortening and imbalance between paretic and active muscles.

Examples of disinhibited primitive reflexes







Positive support reaction



Extensor response

Upper motor neurone syndrome – Negative features

- Weakness (↓ MVC)
  - ↓voluntary activation
  - ↓ motor neurone firing rates

- Slowness of muscle activation and movement
  - Inability to rise to peak force fast enough to generate sufficient power
  - Slowing of muscle relaxation after peak torque

Decreased Force

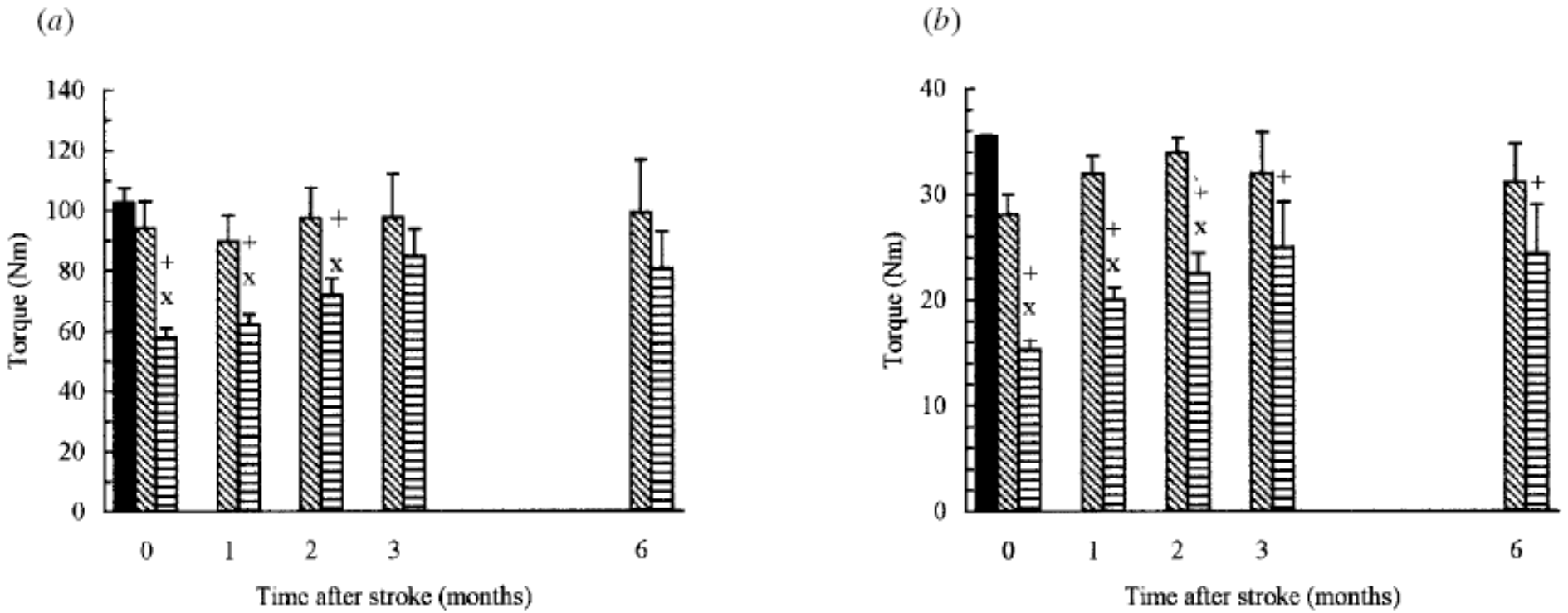
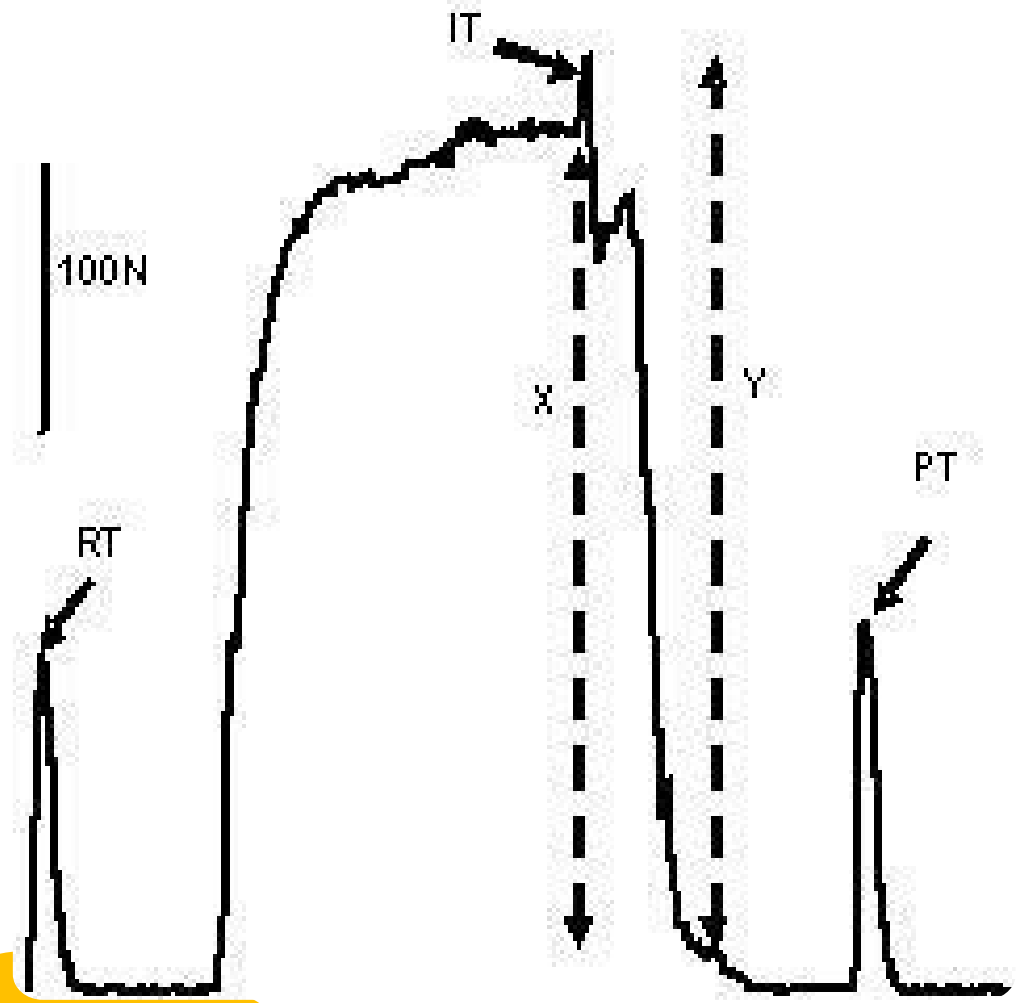


Figure 1 Maximal voluntary torque generation in the quadriceps (a) and hamstring muscles (b). Note different vertical axis scales on the two graphs. Solid bars = control; diagonal stripes = non-paretic; horizontal stripes = paretic. [x] significant differences between paretic and control muscles, [+] significant differences between paretic and non-paretic muscles.

Newham & Hsiao (2001)



Voluntary activation

- Voluntary activation is the neural drive to the motor unit pool supplying a particular muscle.
- Even in healthy individuals, full activation is rare (usually approx. 95%)
- Measuring voluntary activation using the “interpolated twitch” method during maximal voluntary isometric contractions (MVIC)
- Size of the interpolated twitch can be compared with the post-MVIC resting twitch to calculate activation failure

Decreased Voluntary activation

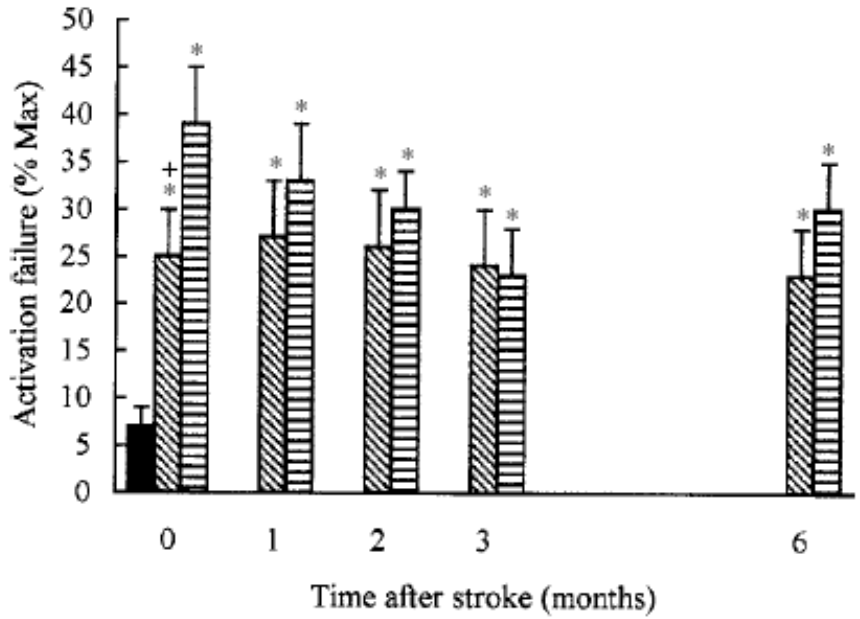


Figure 2 Voluntary activation failure during an isometric maximal voluntary contraction of the quadriceps, determined by twitch superimposition. Both legs of the stroke patients showed greater activation failure than the control subjects throughout the study. Solid bars = control; diagonal stripes = non-paretic; horizontal stripes = paretic. * significant differences between stroke and control muscles, + significant differences between paretic and non-paretic muscles.

- Voluntary activation failure (VAF) was approximately:
  - 7% in control muscles
  - 25% in non-paretic limb after stroke
  - 30-40% in paretic limb after stroke
  - VAF persists for **at least** 6 months
  - VAF contributes to weakness observed after stroke

Decreased motor neuron firing rate

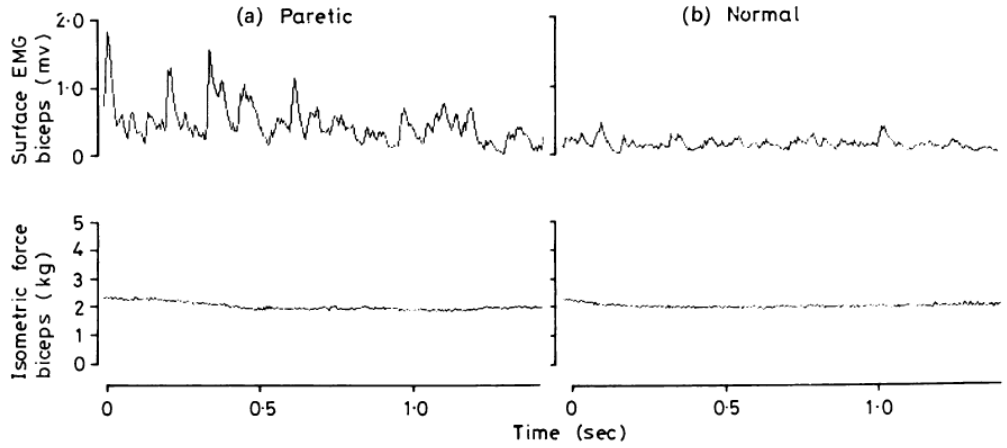
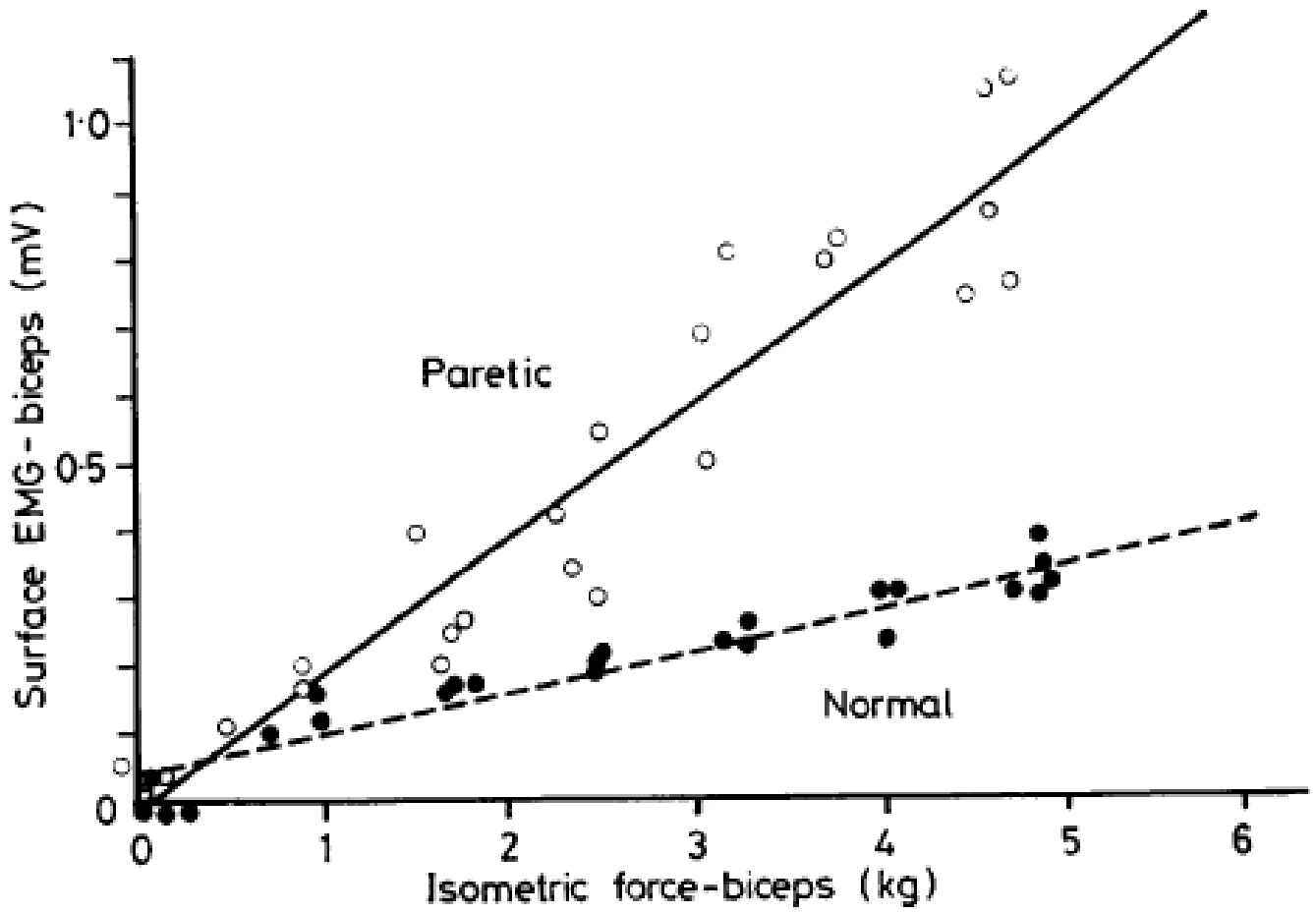


Fig. 2 *Two records of force and surface EMG data taken from the biceps-brachialis (BB) muscle groups of hemiparetic subject 14. Force and EMG were recorded simultaneously and sampled at 5 ms intervals. EMG records shown here were additionally low passed filtered at 50 Hz to prevent aliasing. (a) Taken from paretic muscle group. (b) Taken from normal muscle group.*

- Rectified EMG (rEMG) is proportional to force in healthy muscles
- In paretic muscles after stroke, reduced ability to increase force by increasing firing frequency of motor units results in increased recruitment of a large number and proportion of motor units for same relative force and therefore higher relative rEMG

Decreased motor neurone firing rate



Involuntary Movements

Hemiballismus



Continuous involuntary wide flinging motions of the extremities often involving proximal muscles, on one side of the body -limbs

Chorea



Brief, irregular ‘dance like’ movements that move from one muscle to the next –face, mouth, trunk, limbs

Involuntary movements

Athetosis



slow, involuntary, writhing movements hands and feet

Dystonia



Fluctuating tone in one or more muscles – torticollis, writers' cramp

Learning Outcomes

By the end of this lecture, students should be able to:

- Describe and recognize the clinical presentation of impairments of muscle such as:
  - Hypertonia & hypotonia
  - Rigidity
  - Involuntary movements

- **Identify neural and non-neural factors in spastic hypertonia**

- Understand the aims of physiotherapy intervention in the management of spastic hypertonia

Pathophysiology of spastic hypertonia

Neural component: abnormal enhancement of spinal stretch reflexes

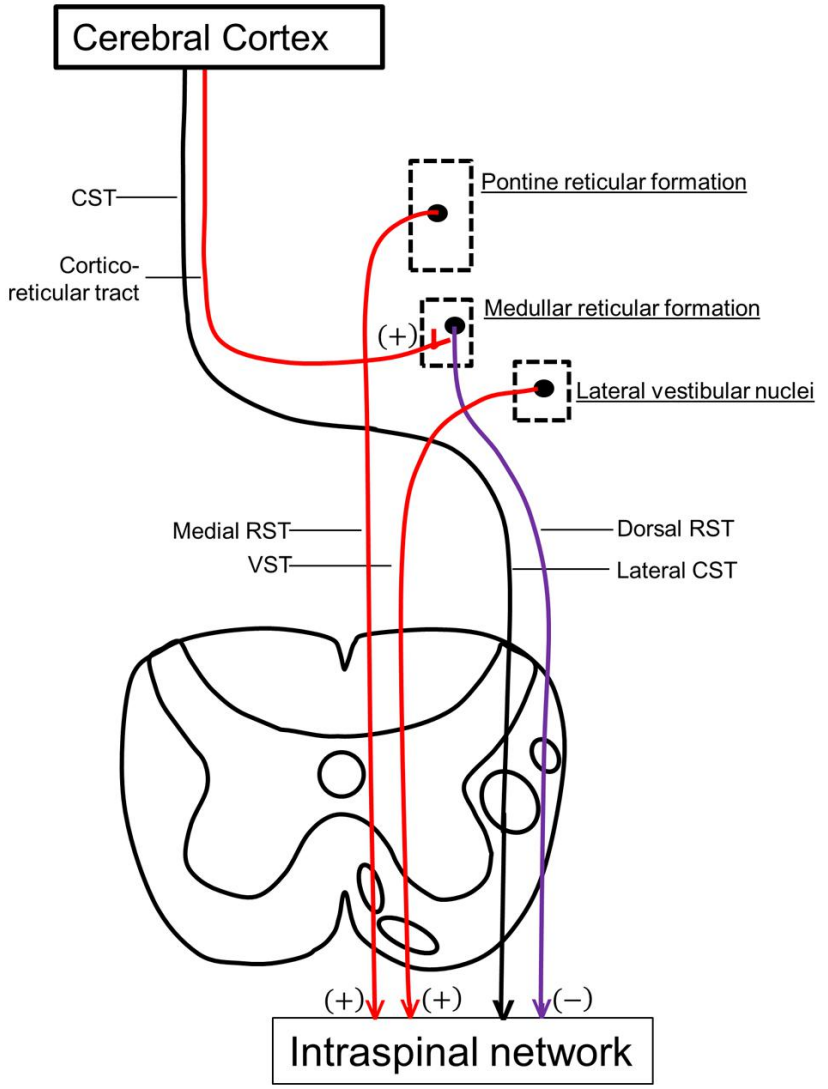
Altered balance between excitatory and inhibitory descending influences on interneurons and motor neurons

Plastic reorganisation in spinal cord

Non-neural component

- Mechanical & morphological adaptive changes to muscle & tendon:
  - cross-bridge connections
  - loss of sarcomeres
  - build-up of connective tissue due to lack of contractile activity
  - muscle fibre properties

Descending influences



Critical features

- Corticoreticular drive to inhibitory area
- Separate inhibitory and excitatory tracts
- Both excitatory and inhibitory tract inhibits flexor reflexes

Illustration of supraspinal control of spinal stretch reflex. CST: cortical spinal tract; RST: reticulospinal tract; VST: vestibular spinal tract; (+): facilitation; (-): inhibition. NOTE: other descending pathways, such as rubrospinal tract, tectospinal tract, medial CST are not shown here.

Main Parapyramidal tracts

DRT Inhibitory System

- Inhibits stretch reflex
- Inhibits flexor reflex

Normal tone:

DRT

Inhibitory effect on SR

MRT/VST Excitatory System

- Facilitates stretch reflex
- Inhibits flexor reflex

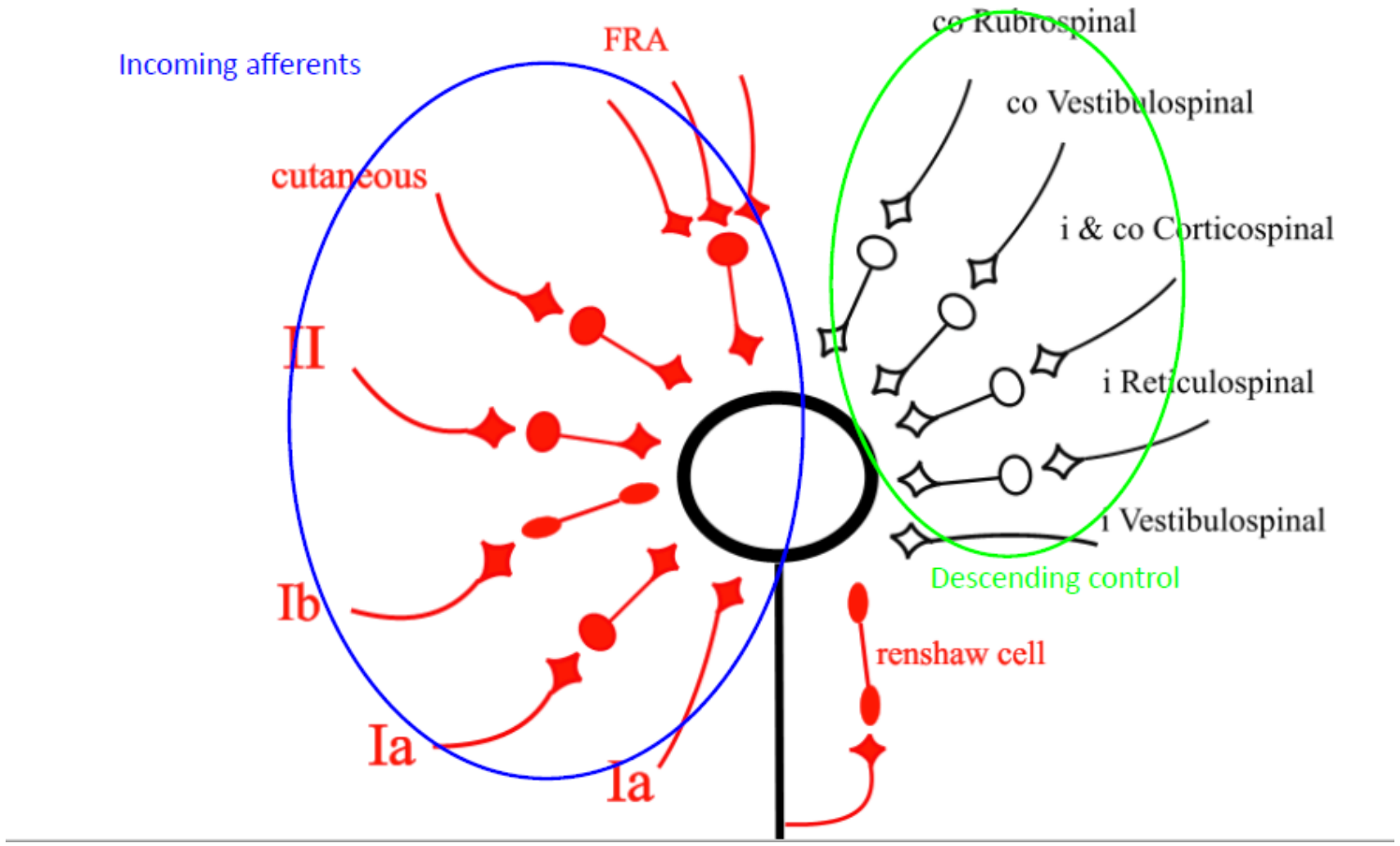
MRT

Facilitatory effect on SR

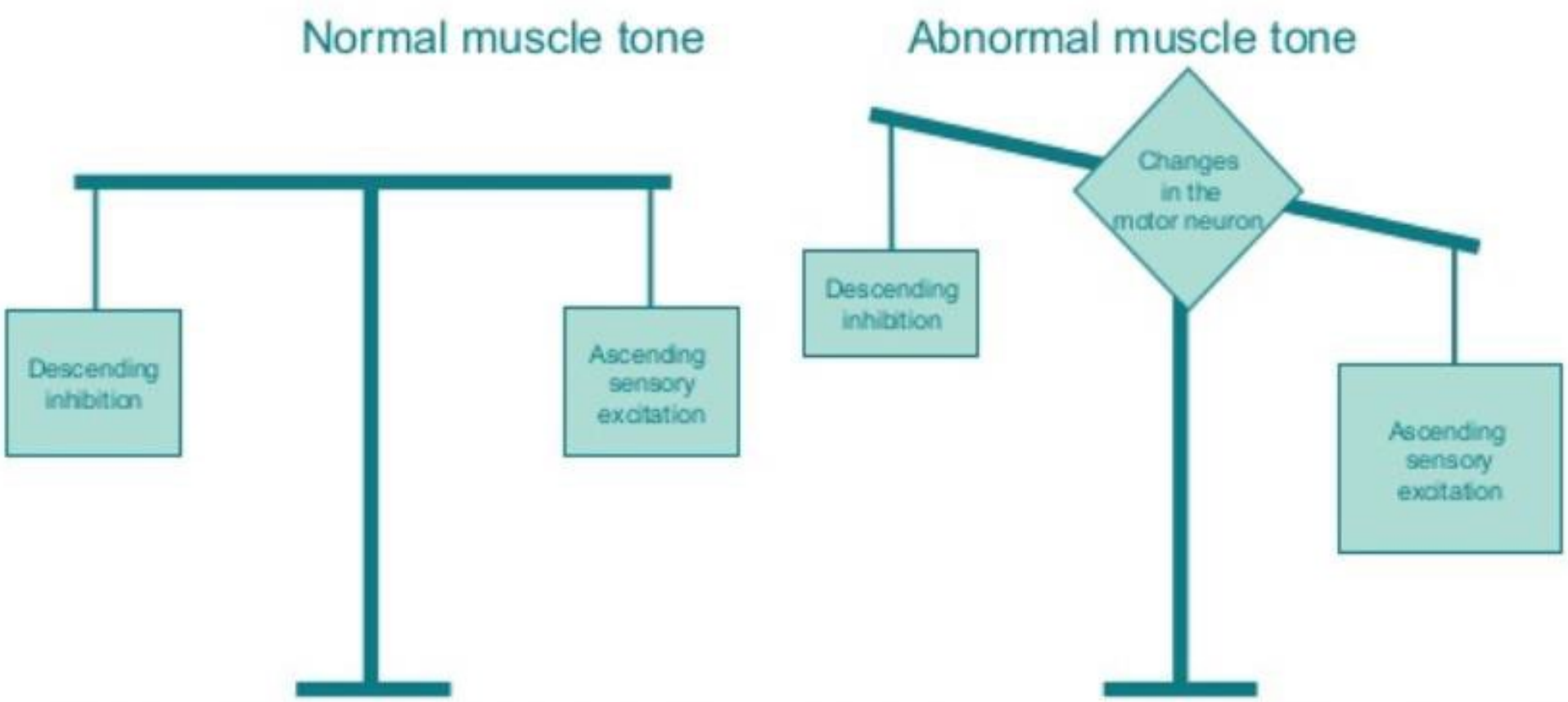
Effect of different lesions

Lesion presentation	Areas Affected	Clinical signs
Cortical (decorticate posturing)	Corticoreticular drive to DRT	↑ stretch reflex Spastic hemiplegia
Incomplete spinal cord (extensor tone)	Lateral CST & DRT	Paresis Severe spasticity
Complete spinal cord	DRT & MRT & VST	Some hypertonus Flexor spasms

Ascending sensory excitation – plastic re-organisation



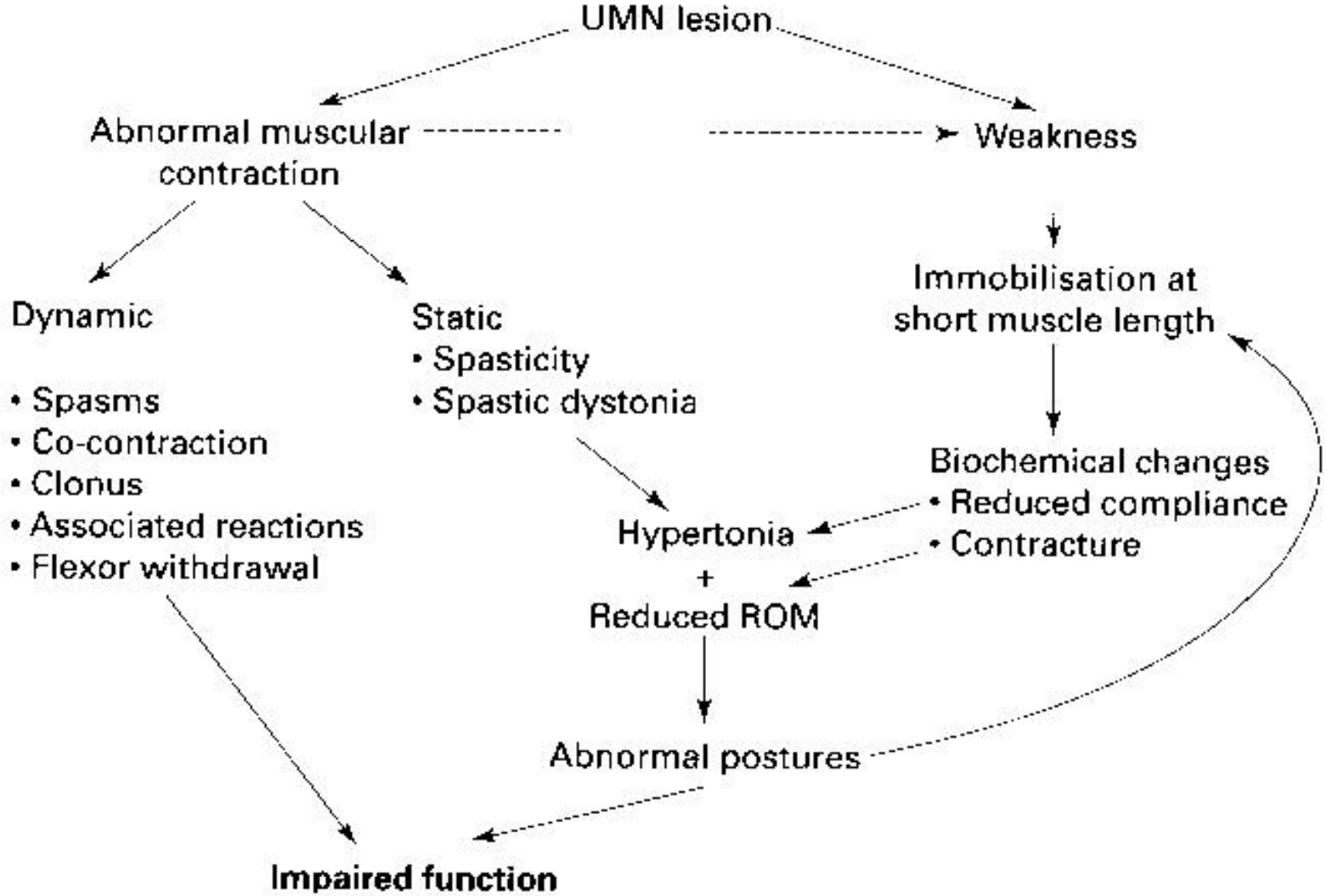
Neural component



Spastic Hypertonia: non neural component

- Not every “tight” muscle is spastic
- Can have increased resistance to passive stretch without spasticity
- Mechanical and morphological adaptive changes:
  - Increased cross-bridge connections during contraction
  - Shortened muscle fires - loss of sarcomeres
  - Build-up of connective tissue due to lack of contractile activity
  - Altered muscle fibre properties

1. Gao F, Zhang LQ. Altered contractile properties of the gastrocnemius muscle poststroke. J Appl Physiol (1985). 2008;105(6):1802 1808.
2. Lieber et al (2004) Structural and functional changes in spastic skeletal muscle. Muscle & Nerve **29** 615 627



A model of the interaction between neural and biomechnical components of hypertonia in the upper motoneurone syndrome.

Learning Outcomes

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- Identify neural and non-neural factors in spastic hypertonia

- **Understand the aims of physiotherapy intervention in the management of spastic hypertonia**

Assessment of spasticity

Resistance to passive movement

↑ with increasing speed of stretch

↑ above a threshold speed or joint angle (“catch”)

varies with muscle length

Scales/Tests

- Modified Ashworth Scale
- Tardieu Scale
- Wartenburg Pendulum test

Modified Ashworth Test

Score	Modified Ashworth Scale
0	No increase in muscle tone
1	Slight increase in muscle tone, manifested by a catch and release or by minimal resistance at the end of the range of motion when the affected part(s) is moved in flexion or extension
1+	Slight increase in muscle tone, manifested by a catch, followed by minimal resistance throughout the remainder (less than half) of the range of movement (ROM)
2	More marked increase in muscle tone through most of the ROM, but affected part(s) easily moved
3	Considerable increase in muscle tone, passive movement difficult
4	Affected part(s) rigid in flexion or extension

doi:10.1371/journal.pone.0126392.t001

Tardieu Scale

Velocity Component

- Velocity of stretch
  - V1 as slow as possible
  - V2 Speed of limb falling with gravity
  - V3 As fast as possible
  - V1 assesses PROM
  - V2 and V3 assess spasticity

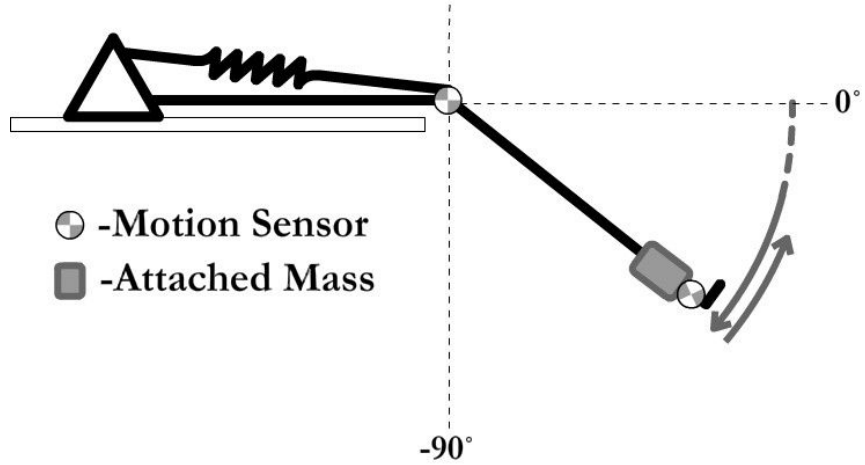
Range component

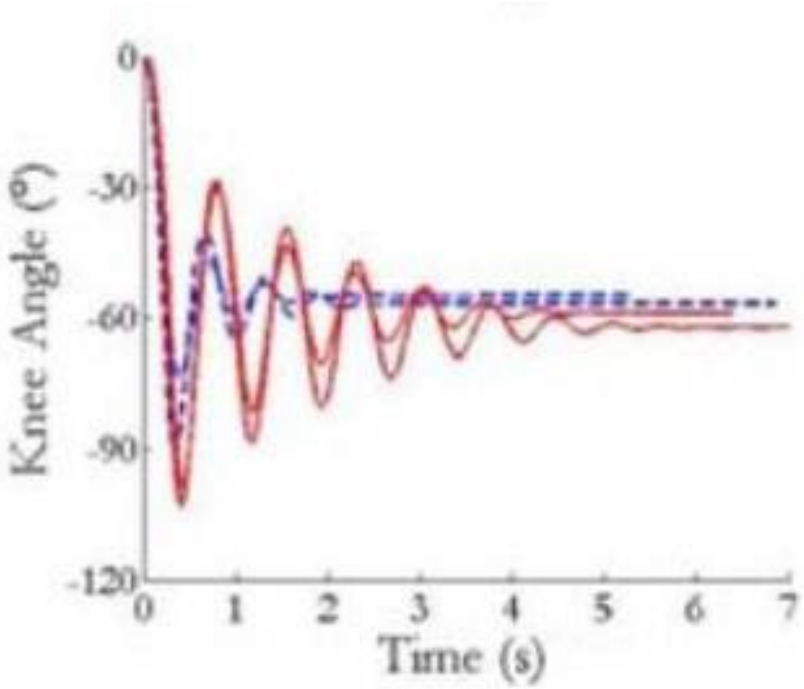
- Range from V1 is full PROM and limits indicate contracture
- The difference between the end of range at V1 (slow) and the angle of resistance/catch at V2 and/or V3 is the spasticity angle.

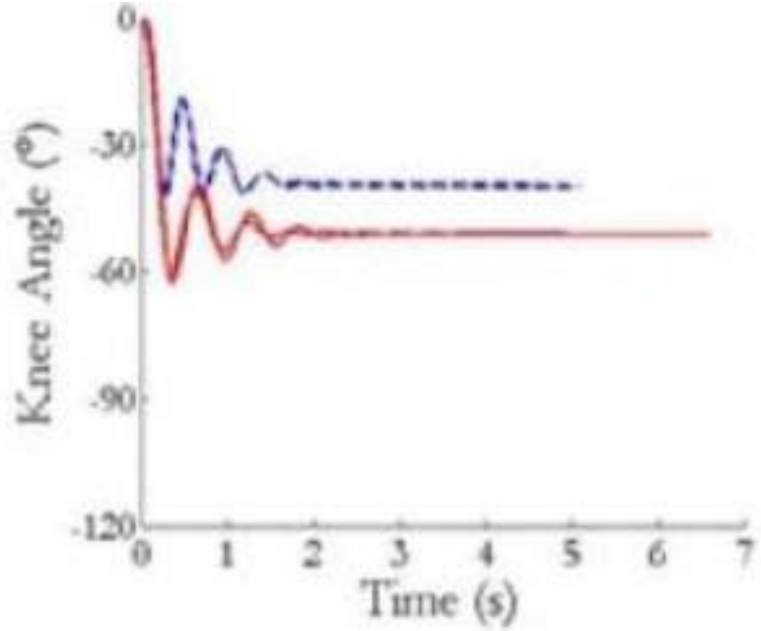
Velocity component

Range component

Wartenburg Pendulum test







Management of spasticity

Consequences



Problems in:
Functioning
Hygiene

Management

- Prevention of aggravating factors
- Education and psychological support
- Physiotherapy
  - Pressure care & positioning
  - Minimise soft tissue changes
  - Optimise function
- Pharmacological
- Surgery

KhederA , and Nair K P S PractNeurol2012;12:289-298

RCP Guidelines: impaired tone - spasticity

- Patients with motor weakness should be assessed for presence of spasticity as:
  - A cause of pain
  - Limiting activities or care
  - Risk factor for contracture

- Interventions should have specific goals and be monitored using appropriate clinical measures
  - Positioning (shown to be effective)
  - Active and/or passive movement
  - Pain control
  - Drugs for persistent troublesome focal (Botulinum toxin)/general (baclofen) spasticity
  - Splinting of arm or hand should **NOT** be used routinely after stroke and splinting and casting should only be offered after individualised specialist assessment and monitoring

Minimising risk of secondary complications (including contracture) RCP guidelines

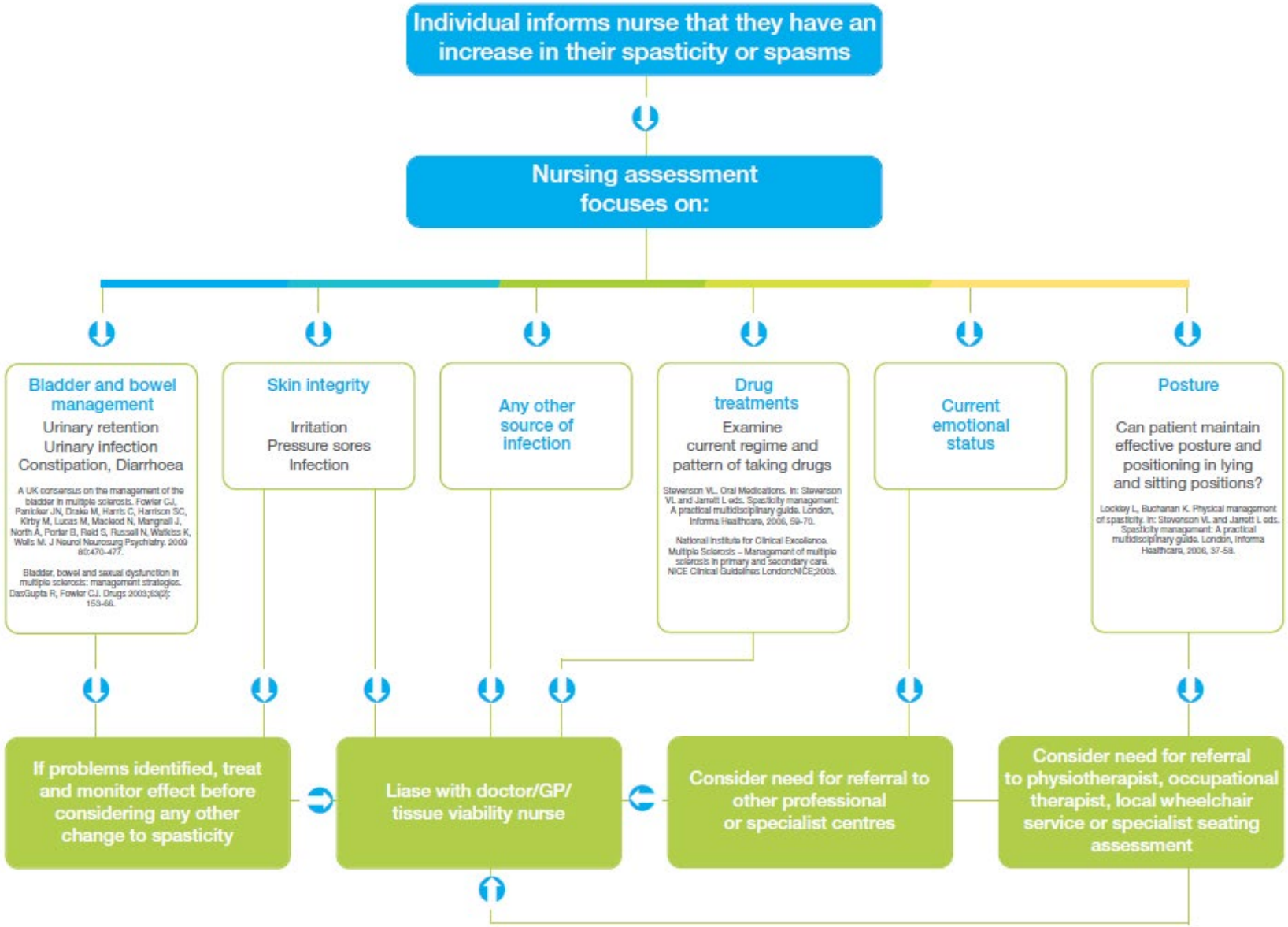
- Patients should have an initial specialist assessment for positioning within 4 hours of arrival in hospital
- Healthcare professionals should be given training on how to position patients correctly after stroke
- When lying and sitting, position in such a way that minimises the risk of complications of aspiration and other respiratory complications, shoulder pain, contractures and skin pressure ulceration
- People with stroke should be helped to sit up as soon as possible

Care Pathway – Spasticity management – MS Trust

- Prevention of sensory stimuli
- The importance of posture
- Moving and handling
- Drug education
- Pain
- Sexual function
- Emotional issues and depression
- Impact on role, employment and social activities

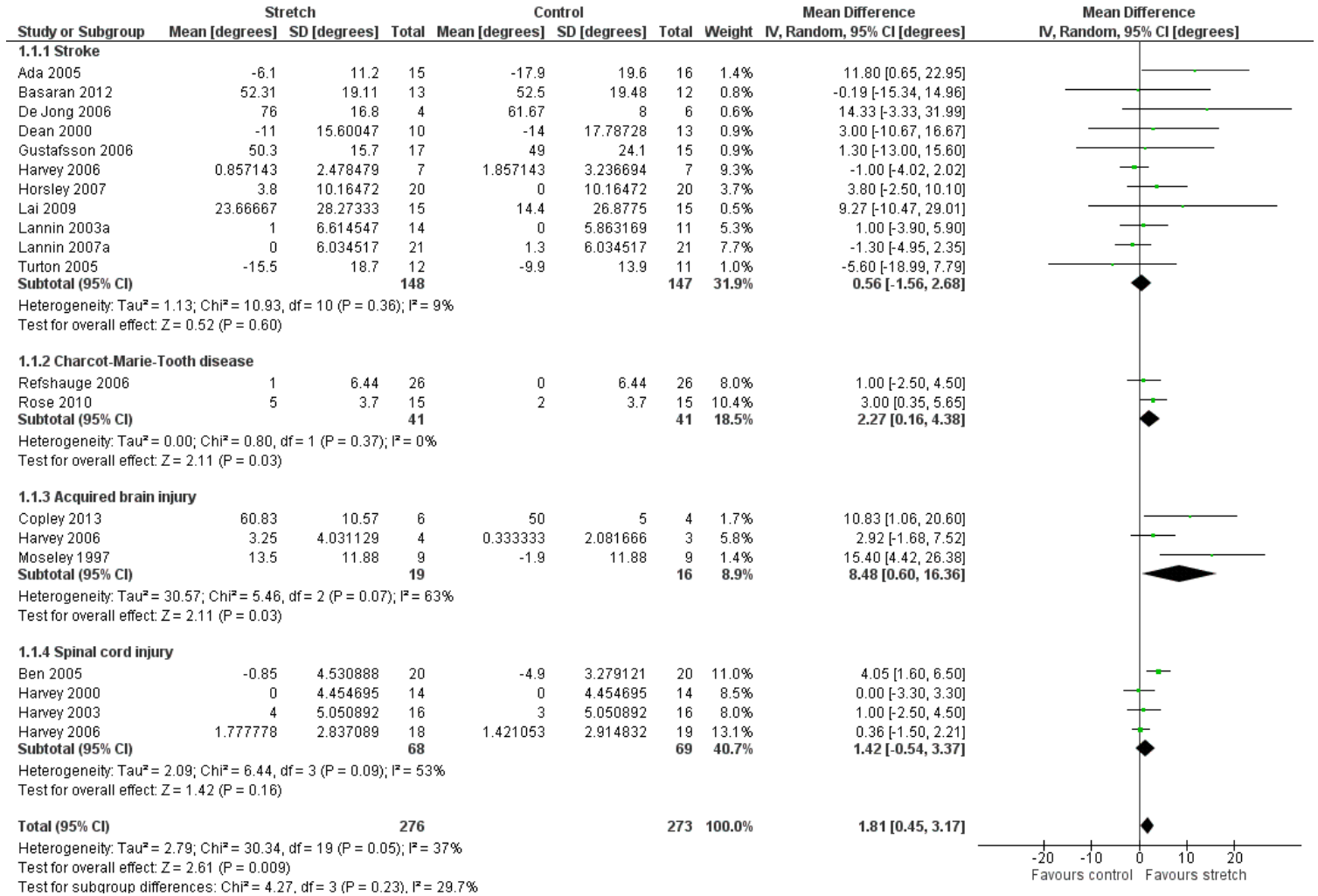
- https://www.mstrust.org.uk/az/spasticity and spasms#what causes spasticity and spasms

Figure 1. Algorithm: Managing an increase in spasticity



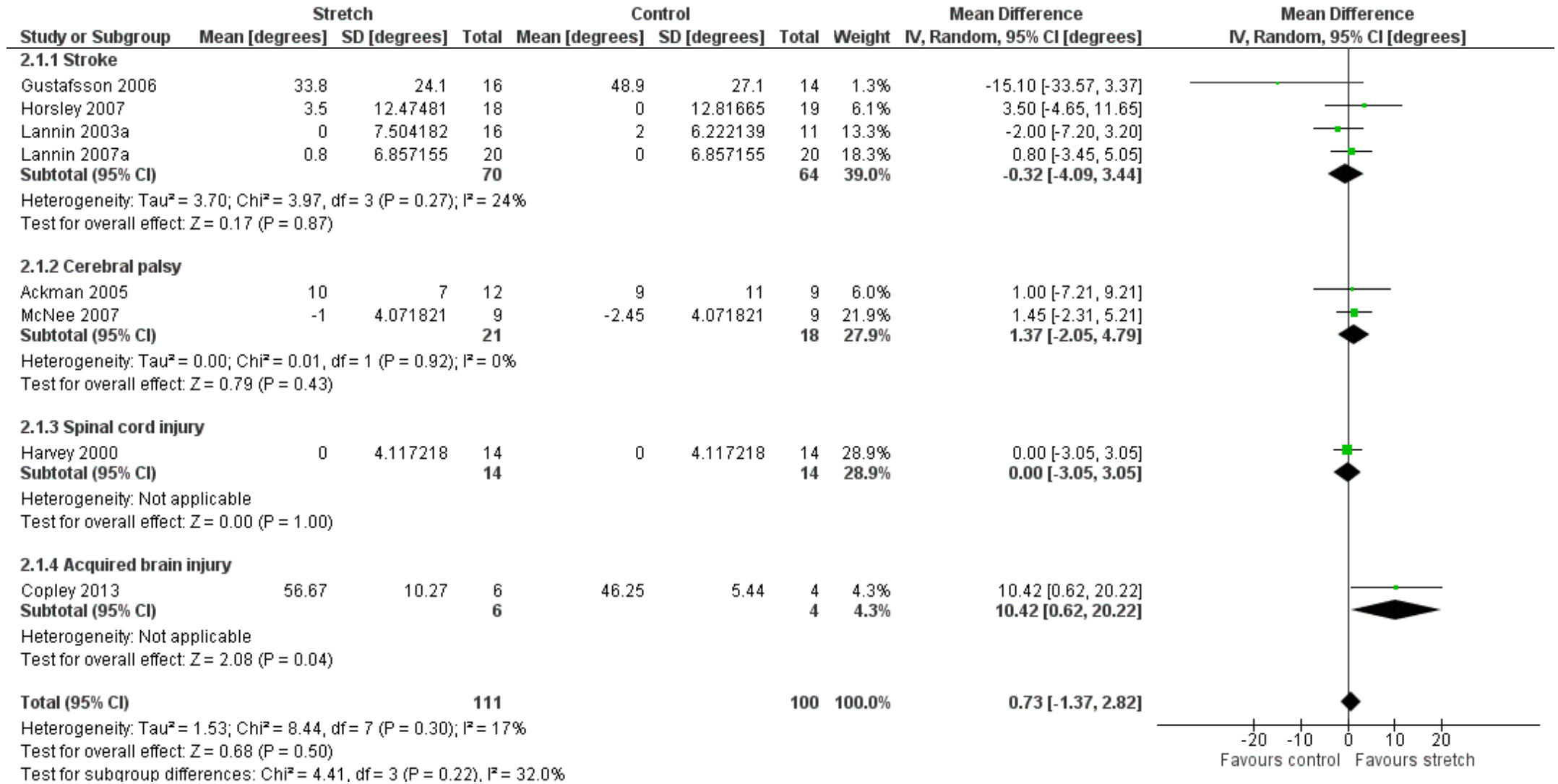
Contracture and stretching

- A large amount of healthcare resources are allocated to the administration of stretch for the treatment and prevention of contractures.
- Contractures are due to both neurally mediated (spasticity directly limiting the extensibility of the muscle-tendon unit) and non-neutrally mediated (structural changes in the muscle-tendon unit and other soft tissue structures overlying joints) factors.
- What is the evidence that stretch may improve joint mobility in contracture?



Forest plot of comparison: Joint mobility - short-term effects following stretch - neurological conditions (degrees)

Harvey LA, Katalinic OM, Herbert RD, Moseley AM, Lannin NA, Schurr K. Stretch for the treatment and prevention of contractures. Cochrane Database of Systematic Reviews 2017, 1. Art. No.: CD007455. DOI: http://dx.doi.org/10.1002/14651858.CD007455.pub3



Forest plot of comparison: Joint mobility - long-term effects following stretch - neurological conditions (degrees)

Harvey LA, Katalinic OM, Herbert RD, Moseley AM, Lannin NA, Schurr K. Stretch for the treatment and prevention of contractures. Cochrane Database of Systematic Reviews 2017, 1. Art. No.: CD007455. DOI: http://dx.doi.org/10.1002/14651858.CD007455.pub3

Conclusions on the role of stretching on contractures

- The results of this systematic review are sufficiently robust to indicate that stretch, as typically provided by physiotherapists, **does not produce clinically meaningful effects on severity of contractures** in people with neurological or non-neurological conditions.

- We do not recommend further studies looking at the short-term effects of stretch on joint mobility in either people with neurological or non-neurological conditions because the quality of evidence indicating that stretch is ineffective is high and further studies are unlikely to change these findings.

- While the quality of evidence about the long-term effects is less rigorous, there is no theoretical basis upon which to believe that stretch may have long-term effects on joint mobility in the absence of a short-term effect.

- There may be worth in examining the effectiveness of stretch administered with motor training or botulinum toxin in people with neurological conditions.

- There may also be worth in specifically looking at the effectiveness of stretch for the prevention of contracture in those at high risk of developing contracture (e.g. people with traumatic brain injury).

Orthoses and serial casting



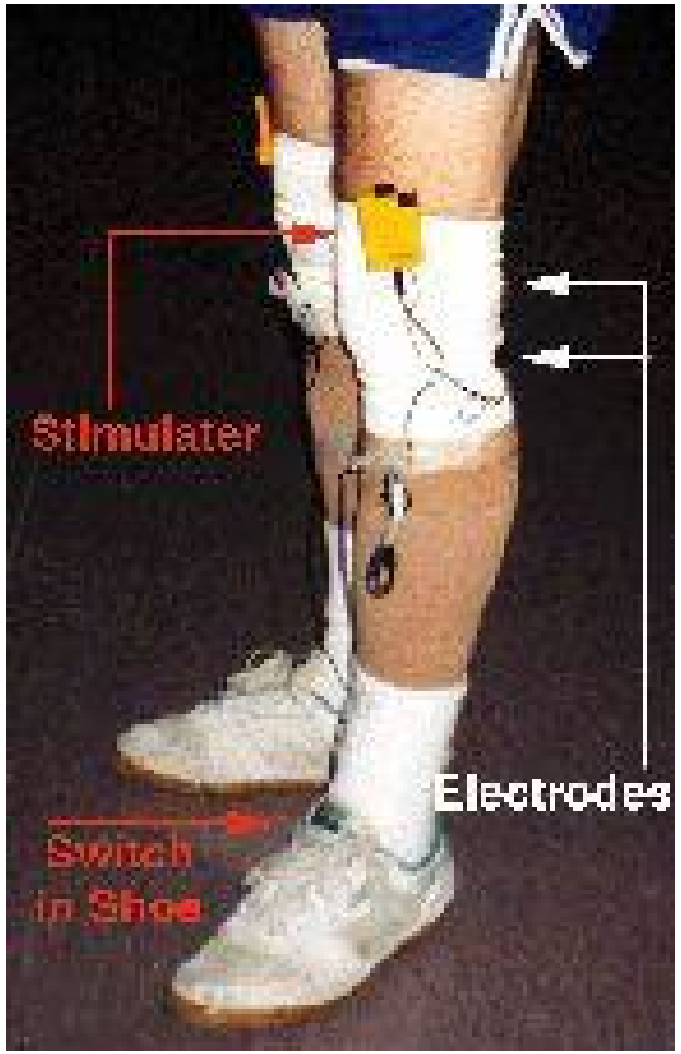


- STANDING: maintaining muscle length, modulating tone, activating extensor activity.
- Can be carried out while patient still unconscious
- Should be dynamic





Functional Electrical Stimulation





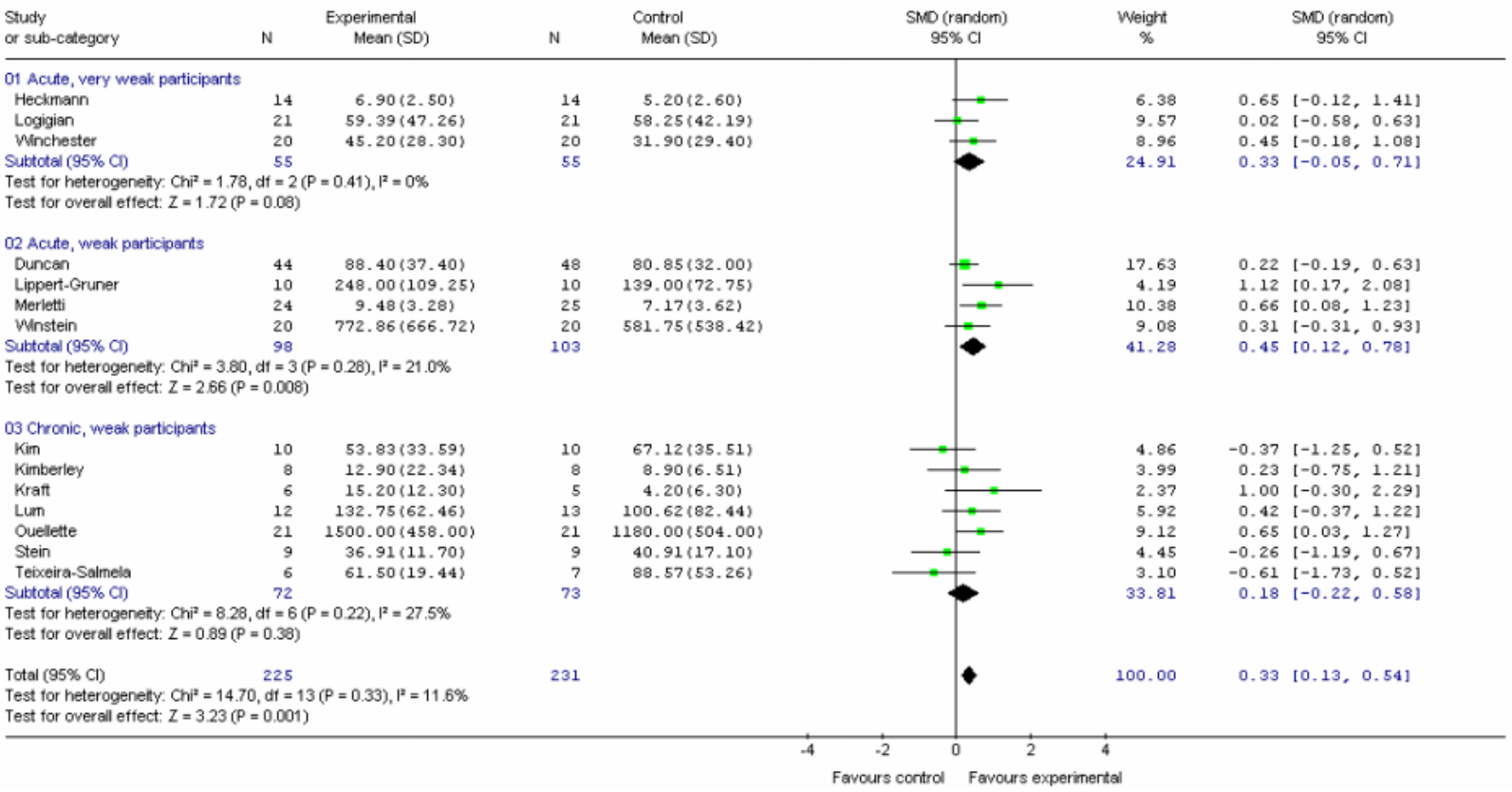
FES: to aid walking in patient with foot drop

Spasticity and resistance exercise

- Ada L, DorschS, Canning CG. Strengthening interventions increase strength and improve activity after stroke: a systematic review. AustJ Physiother. 2006;52(4):241-8.
- Silva Borges, CA., GastaoKC., et al. 2009 Effect of resisted exercise on muscular strength, spasticity and functionality in chronic hemiparetic subjects: a systematic review. J App Res 9, (4) 147-158.
- Merino-Andrés J, García de Mateos-López A, Damiano DL, Sánchez-Sierra A. Effect of muscle strength training in children and adolescents with spastic cerebral palsy: A systematic review and meta-analysis. Clinical Rehabilitation. 2022;36(1):4-14.

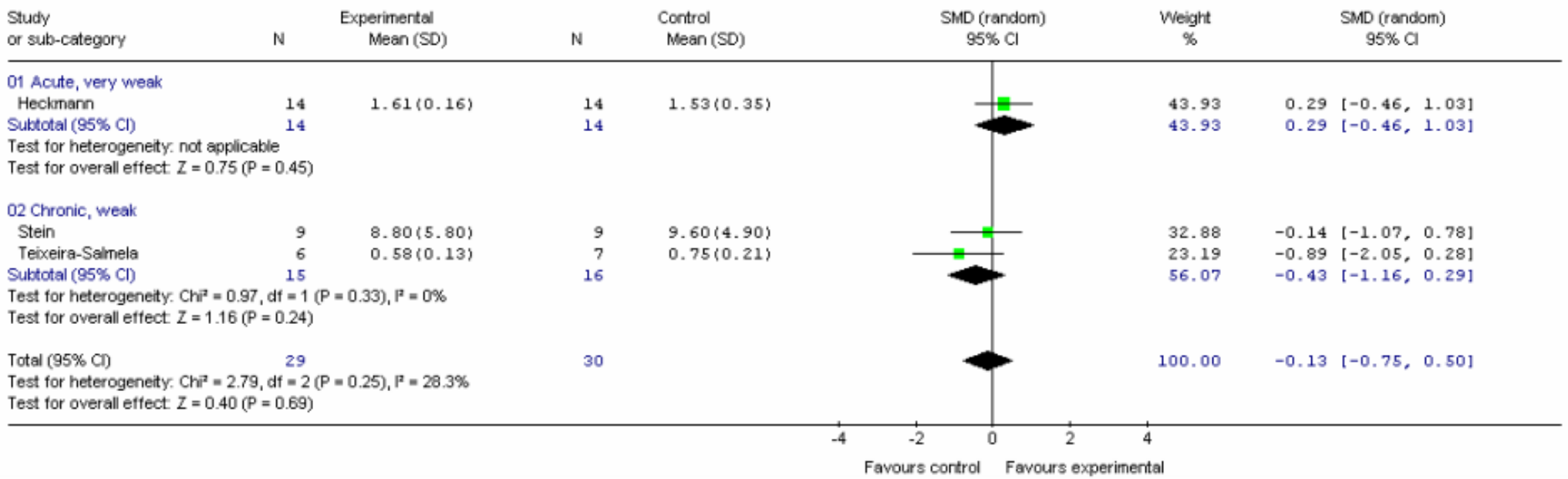
- Concern regarding “associated reactions” and increased spasticity due to effortful activity
- Effortful voluntary contractions in one part of the body, eliciting increased muscle activity or spasticity in affected limbs elsewhere in the body. E.g increase in upper limb flexor tone on sit to stand in patient with upper limb spasticity following stroke

Figure 2. Examination of the effect of strengthening interventions on strength by pooling post-intervention data from 14 trials.



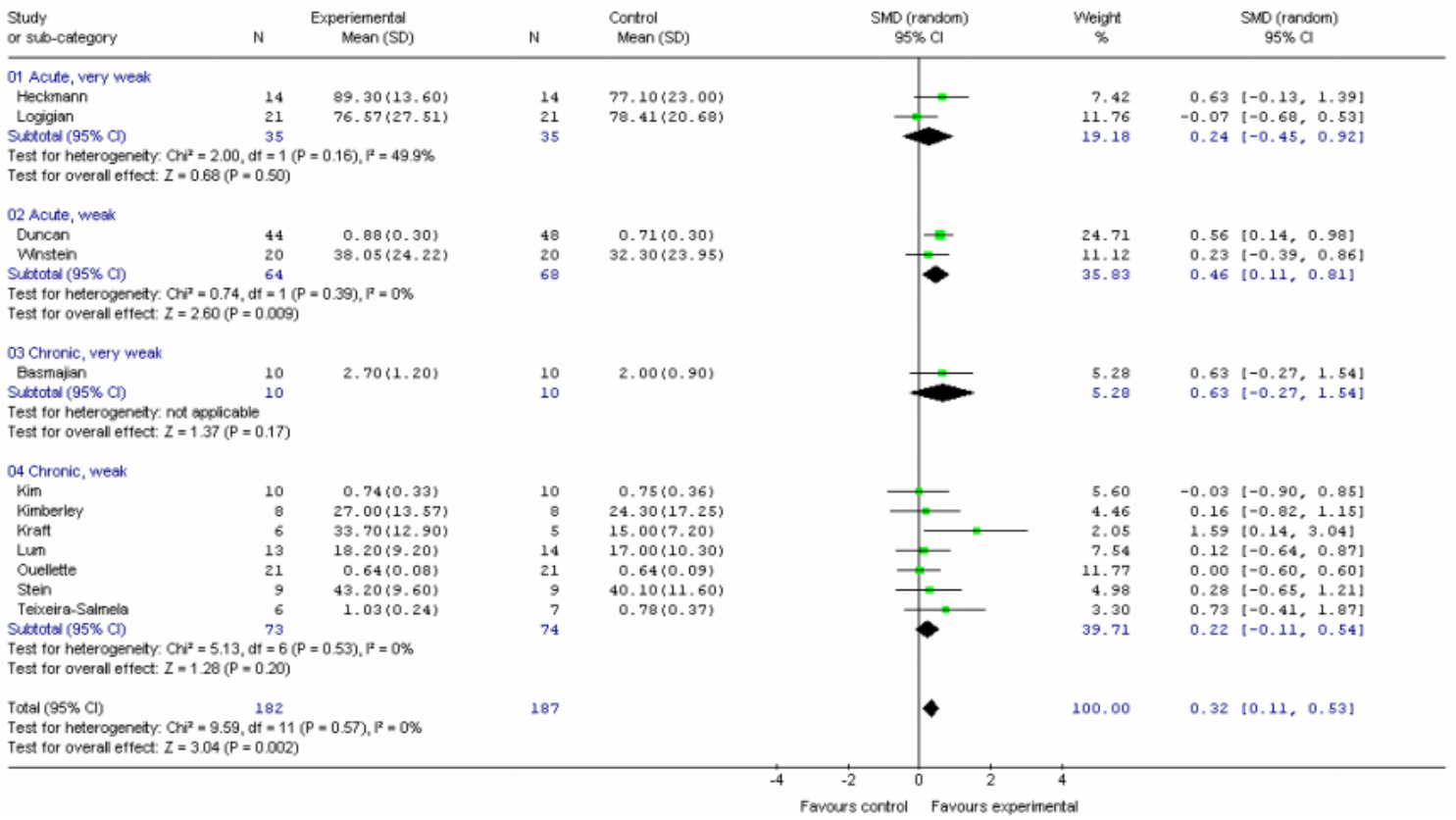
Ada et al (2006) Strengthening interventions increase strength and improve activity after stroke: a systematic review Aus J Physiotherapy 52 241

Figure 4. Examination of the effect of strengthening interventions on spasticity by pooling post-intervention data from 3 trials.



Strengthening does not increase spasticity

Figure 6. Examination of the effect of strengthening interventions on activity by pooling post-intervention data from 12 trials.



Strengthening exercise improves function



Non-pharmacological interventions for spasticity

- Overview of systematic reviews (18 SRs)

- **Moderate quality evidence** for:
  - electro-neuromuscular stimulation and
  - electro-acupuncture as an adjunct to routine care (pharmacological and rehab)

- **Low quality** evidence for
  - CIMT, stretch, dynamic elbow splinting, OT in stroke
  - Extracorporeal Shock Wave Therapy (ESWT), transcranial direct stimulation (TDS) and transcutaneous electrical nerve stimulation (TNS) in other neurological conditions
  - Physical activity programmes, repetitive nerve stimulation in Multiple Sclerosis
  - Vibration therapy for spinal cord injury (SCI)
  - Stretching for other neurological conditions

Khan et al (2019). Ann Phys Rehabil Med 62 (4) 265-273

Drugs for spasticity

- **Troublesome focal spasticity –Botulinum Toxin (phenol)**
- **Botulinum - acts at neuromuscular junction by inhibiting Ach release and paralysing muscle.**
- **Carried out with splinting and/or rehab and reassessed 3-4 months later.**
- Side effects:
  - Injection site pain
  - haematoma,
  - muscle necrosis,
  - phlebitis

- **Troublesome generalised spasticity – baclofen (tizanidine, cannabinoids)**
- **Baclofen - acts centrally to reduce the release of excitatory neurotransmitters (Glutamate)**
- ***Side effects baclofen***
  - ***Oral:*** drowsiness, confusion, headache, lethargy;
  - ***Intrathecal:*** pump malfunction
- **Side effects tizanidine**
  - Dizziness,
  - sedation,
  - dry mouth
- **Diazepam acts to increase levels of inhibitory neurotransmitter**
- **Dantrolene acts within muscle to reduce Ca2+ release for contraction**



References

1. Graham L (2013). Management of spasticity revisited Age and Ageing 42 435-441 http://ageing.oxfordjournals.org/content/42/4/435.full.pdf+html

2. Khederand Nair (2012). Spasticity: pathophysiology evaluation and management Practical Neurology 12 289-298 http://pn.bmj.com/content/12/5/289.full.pdf+html

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Weblinks to guidelines

- http://www.nice.org.uk/guidance/index.jsp?action=byTopic&o=7208
- http://www.rcplondon.ac.uk/sites/default/files/national-clinical-guidelines-for-stroke-fourth-edition.pdf